

Genetic Risk Modeling for Schizophrenia Using Public GWAS and 1000 Genomes Data

Sahil , bachelors in Mechanical Engineering, New Delhi

Date: 16 ,March. 2026

GitHub: <https://github.com/mmeridius544-cyber/Genetics-Influence-On-pschiatric-deceases>

Abstract.

Schizophrenia is a highly polygenic psychiatric disorder with substantial heritability, but translating genome-wide association study (GWAS) findings into individual-level risk models remains challenging. In this project, I implemented an end-to-end computational pipeline that integrates summary-level schizophrenia GWAS results from the Psychiatric Genomics Consortium with individual-level genotype data from the 1000 Genomes Project to model genetic risk. The pipeline performs variant-level quality control and harmonization, including allele matching, ambiguous SNP filtering, Hardy–Weinberg equilibrium checks, minor allele frequency thresholds, and explicit X-chromosome handling. Using this framework, I compute per-variant association statistics and polygenic scores, and generate standard GWAS diagnostics such as Manhattan and QQ plots, ancestry principal component analysis (PCA), and polygenic risk score distributions. The results recover polygenic signal across the genome and demonstrate that even with publicly available cohorts (Reason Why the plots are bit inflated) it is possible to build a reproducible “SNP layer” suitable for extension to larger datasets (e.g., UK Biobank) and multimodal models. This summary describes the data, methods, and key findings as a basis for future supervised work in psychiatric genetics and machine learning.

Data and Methods

Data

- GWAS summary statistics: Schizophrenia GWAS results from the Psychiatric Genomics Consortium (PGC), including per-SNP effect sizes and p-values.
- Individual genotypes: 1000 Genomes Project phase 3 data (VCF/PLINK) from their VCF file directory repository, covering autosomes and chromosome X across diverse ancestries.

Pre-processing & quality control(QC)

- Harmonized alleles between PGC summary statistics and 1000 Genomes (reference/alternate strands, resolving ambiguous A/T and C/G SNPs where necessary).
- Applied variant-level QC: minor allele frequency and allele count filters, Hardy–Weinberg equilibrium checks, and dosage mean/variance sanity checks.
- Implemented explicit X-chromosome handling: distinguished PAR and non-PAR regions, inferred sex from X-chromosome heterozygosity, and modeled non-PAR X in diploid space for consistency with autosomes.
- Derived ancestry principal components from 1000 Genomes genotypes to visualize population structure and enable adjustment for stratification.

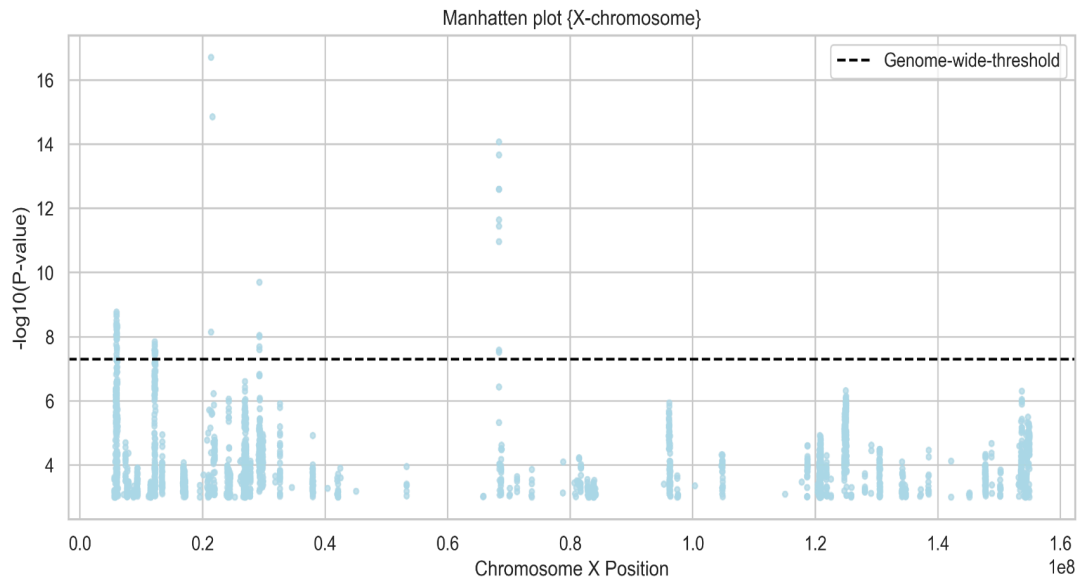
Association & risk modeling

- Merged GWAS effect sizes with harmonized individual-level genotypes to compute per-variant contributions to liability.
- Constructed polygenic scores (PRS) as weighted sums of risk alleles across genome-wide SNPs, experimenting with different p-value thresholds for SNP inclusion.
- Implemented all steps in a single Jupyter notebook with configurable paths and parameters to support extension to other phenotypes or cohorts.

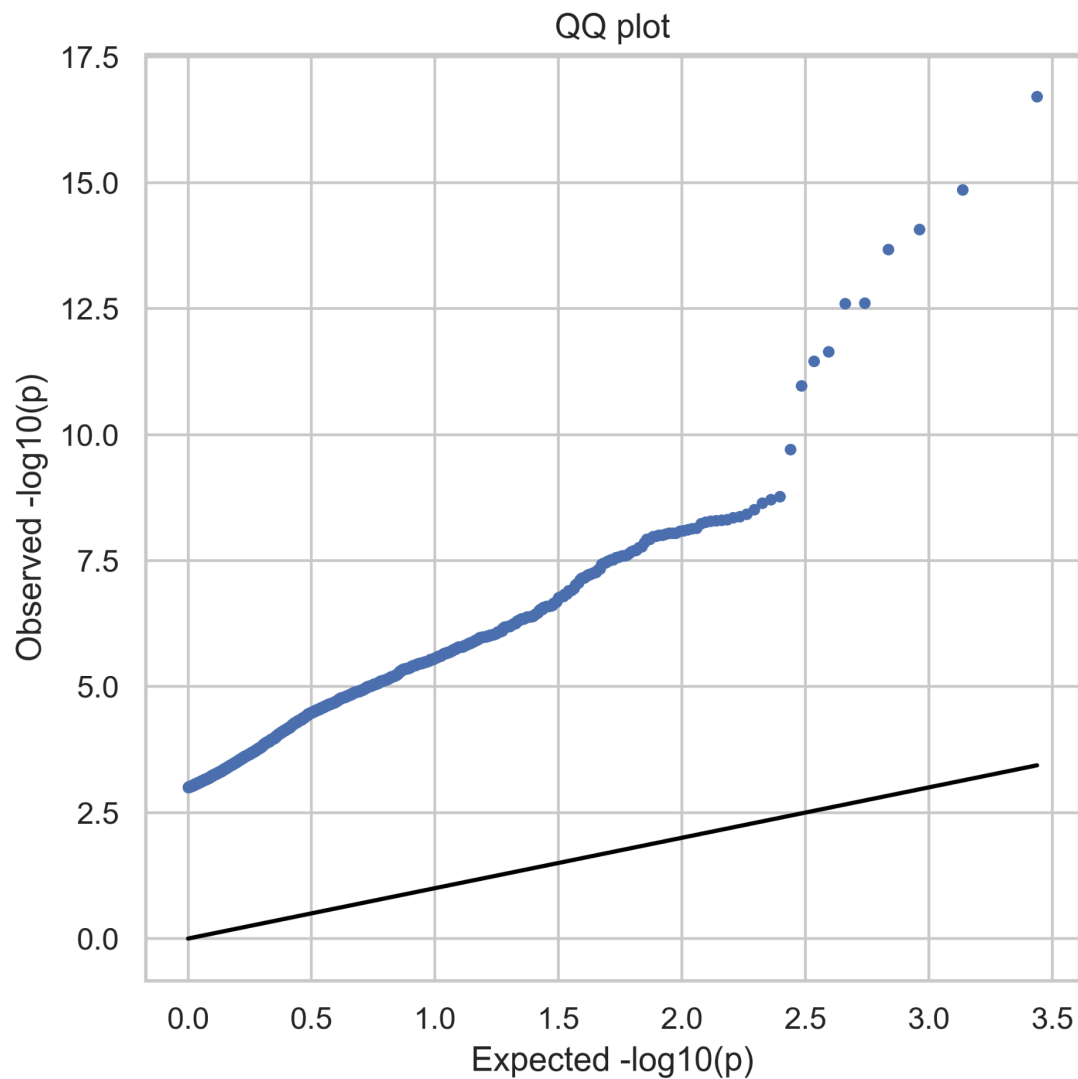
Results

Genome-wide signal. The Manhattan plot shows widespread polygenic signals across the genome, with multiple peaks reaching genome-wide significance thresholds and clusters of associated variants in regions previously implicated in schizophrenia and neuronal function. This indicates that the harmonization and QC pipeline correctly

recovers known association structure rather than generating spurious artifacts.

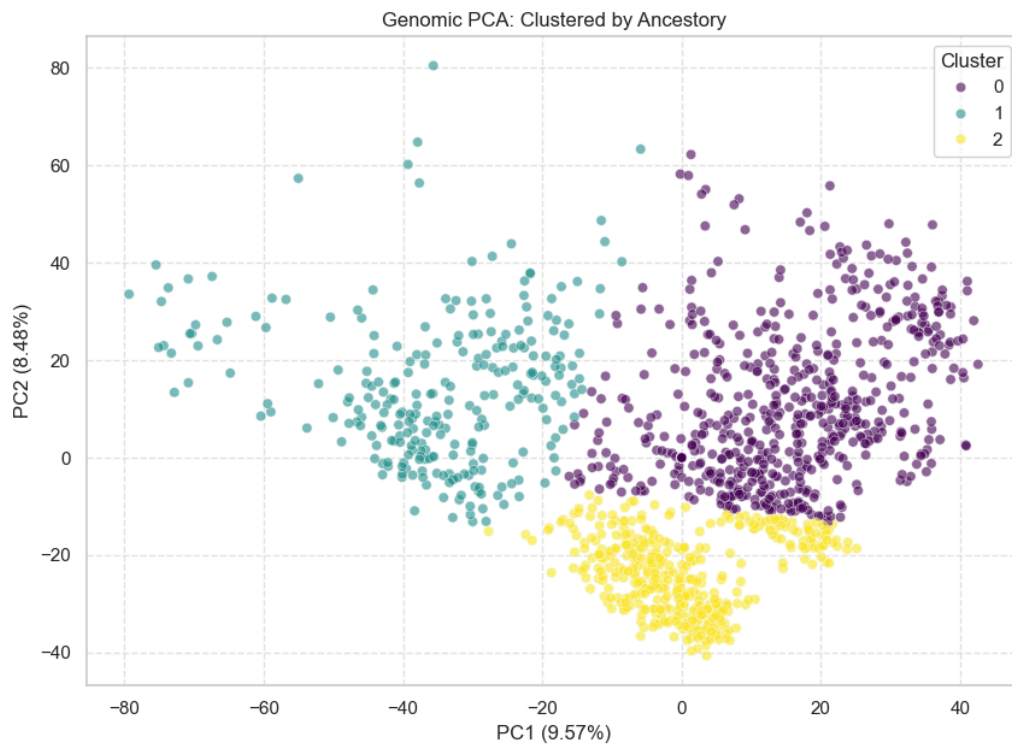


Calibration and polygenicity. The QQ plot displays a controlled deviation from the null expectation at the tail, consistent with true polygenic enrichment and modest inflation compatible with complex trait architecture rather than uncorrected population stratification. This pattern is in line with previous observations that schizophrenia is influenced by many common variants of small effect.



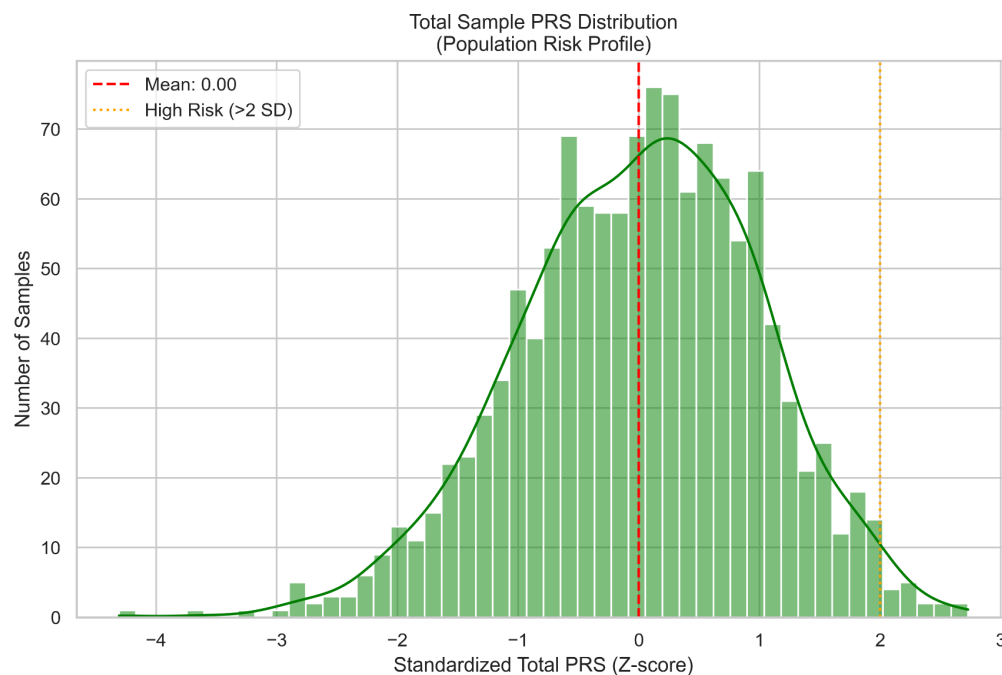
Ancestry structure. PCA of 1000 Genomes genotypes reveals clear clustering of individuals into major continental ancestry groups, confirming that the pipeline captures the expected population structure. These principal components can be used

as covariates in downstream association or prediction models to reduce confounding by ancestry.



Polygenic risk distributions. Polygenic risk scores, constructed by aggregating GWAS effect-weighted alleles, show a spread of liability across individuals with tails corresponding to higher and lower genetic risk. Although this pilot implementation uses public data rather than clinically ascertained cases and controls, the resulting distributions demonstrate that the framework can be used to stratify individuals by

genetic risk and could be plugged into machine-learning models in richer cohorts such as UK Biobank.



Drawbacks.

- The current implementation relies on publicly available cohorts (PGC summary statistics and 1000 Genomes) rather than large controlled datasets such as UK Biobank or full individual-level PGC schizophrenia cohorts, which limits statistical power and clinical realism.
 - The model is genetics-only and focuses on common SNPs; it does not yet incorporate rare variants, structural variation, gene–gene interactions, or environmental risk factors.
 - No neuroimaging, transcriptomic, or clinical/behavioral data are included at this stage, so the pipeline cannot yet capture multimodal signatures of schizophrenia.
- Future work

Future ambitious work

- Extend the pipeline to larger controlled datasets such as UK Biobank to obtain more reliable risk estimates and to evaluate machine learning models beyond simple regression.
- Integrate additional modalities (e.g., structural and functional neuroimaging, transcriptomic data) to build multimodal models that link genetic risk to brain-level and molecular phenotypes.
- Explore advanced ML architectures (e.g., regularized models, gradient boosting, and multimodal fusion networks) and compare them to classical PRS-based approaches for schizophrenia and related psychiatric disorders.

Full code, logs, and plots are available at ;,

<https://github.com/mmeridius544-cyber/Genetics-Influence-On-pschiatric-deceases>